

Hybrid Scene Generation Benchmark

Genere le 2026-08-02 16:19:10.

Ce document audite la nouvelle chaine runtime :

- room brief source
- commande cardinale
- metadata JSON produite par le LLM local
- compilation hybride deterministe SpatialState -> .scene
- audit du .scene compile et rendu PNG

Comparaison :

- le benchmark historique SCENE_GENERATION_BENCHMARK.md mesure la voie directe brief -> .scene brut par Ministral
- ce benchmark mesure la voie runtime brief source -> room JSON -> compilateur hybride -> rendu

Parametres :

- rendu : 800x400, 8 spp, 3 bounces, 1 direct samples
- runtime LLM : temperature=0.0, n_predict=1024, json_grammar=off

Resultat global : 10/10 rendus valides, dont 0 avec fallback.

Case	Dir	Status	Metadata	Scene	Infer ms	Prep ms	Render ms	Triangles
entry_gate_dusk	NORTH	valid	llm	hybrid	7787.57	10898.24	803.02	460
sally_port_checkpoint	NORTH	valid	llm	hybrid	9338.74	12441.81	645.94	628
badge_vestibule	NORTH	valid	llm	hybrid	7549.01	10661.65	607.01	340
server_aisles_dense	NORTH	valid	llm	hybrid	9187.81	12328.85	1330.60	1062
control_hub	NORTH	valid	llm	hybrid	7077.20	10168.62	560.99	424
backup_vault	EAST	valid	llm	hybrid	8082.58	11261.46	668.25	676
cooling_exchange_bay	NORTH	valid	llm	hybrid	8102.99	11222.40	890.40	520
loading_dock_dust	EAST	valid	llm	hybrid	9683.33	12801.16	607.95	436
roof_watch_dusk	NORTH	valid	llm	hybrid	7900.49	11002.84	1176.02	894
stairhead_parapet	NORTH	valid	llm	hybrid	8223.88	11470.38	701.68	508

Entry Gate

A concrete gate court faces the datacenter threshold. Dust clings to the bars and the badge reader blinks beside a service intercom.

Direction benchmarkee : north Status : valid Updated place : Access Tunnel Intent : move_north_generated_room Metadata source : llm Scene source : hybrid Rendered scene source : generated_room_cache Turn fallback : not needed Prompt dump : 89.78 ms Command wall time : 11831.36 ms

Inference time : 7787.57 ms Preparation time : 10898.24 ms Compile estimate : 3110.67 ms Render time : 803.02 ms Prompt tokens : 798 Generated tokens : 319 Triangles : 460 Materials : 40 Scene lines : 18 Scene chars : 1402

Origin state :

generated/hybrid_scene_generation_benchmark/states/entry_gate_dusk.origin.session.json Result state : generated/hybrid_scene_generation_benchmark/states/entry_gate_dusk.result.session.json

Current generated room :

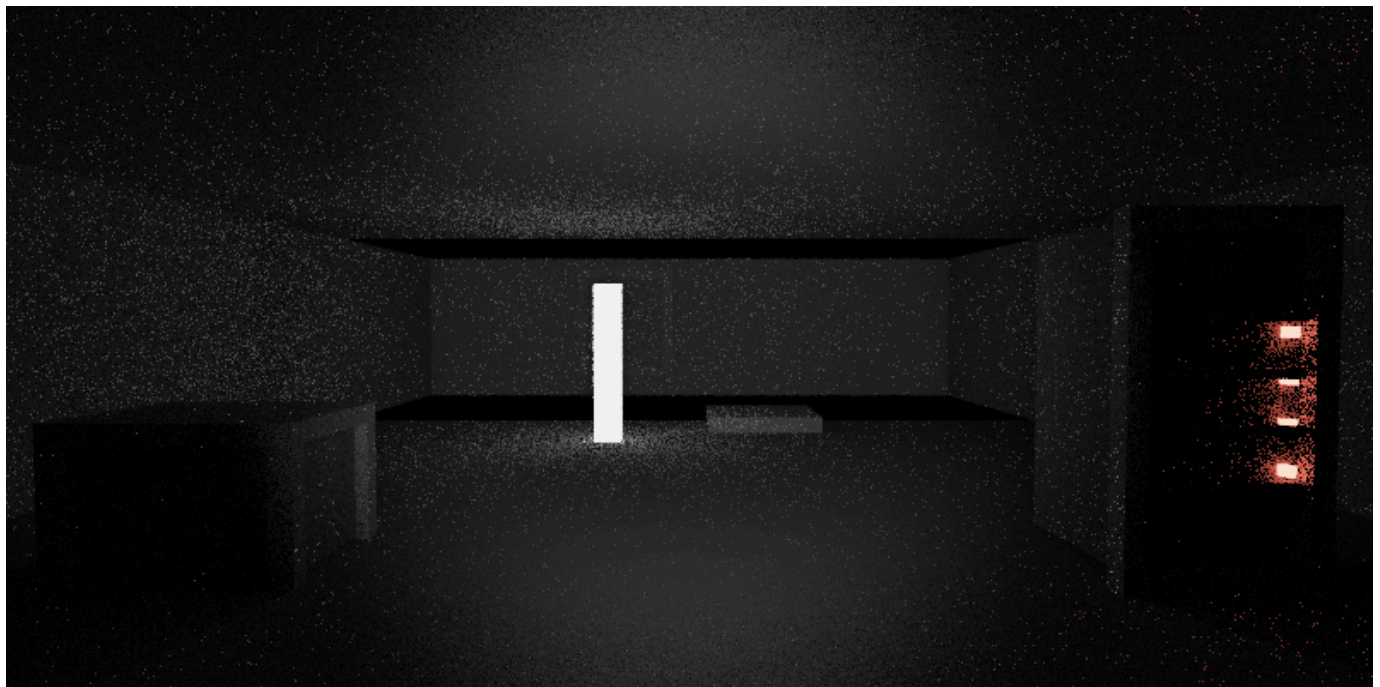
generated/hybrid_scene_generation_benchmark/states/entry_gate_dusk.generated_room.json Prompt : generated/hybrid_scene_generation_benchmark/prompts/entry_gate_dusk.prompt.txt Run stdout : generated/hybrid_scene_generation_benchmark/logs/entry_gate_dusk.stdout.txt Run stderr : generated/hybrid_scene_generation_benchmark/logs/entry_gate_dusk.stderr.txt Raw room JSON : generated/hybrid_scene_generation_benchmark/responses/entry_gate_dusk.raw.txt Normalized room JSON :

generated/hybrid_scene_generation_benchmark/responses/entry_gate_dusk.generated_room.json

Compiled scene : generated/hybrid_scene_generation_benchmark/scenes/entry_gate_dusk.scene

Source room brief :

- location_id: gate
- location_archetype: threshold courtyard
- time_of_day: dusk
- visibility_level: clear
- desert_state: still
- interior_density: sparse
- alert_level: 1
- blocked_exits: west, south



Sally Port Checkpoint

A narrow checkpoint sits between two heavy gates. A code keypad, a dead camera head, and a steel cabinet make the space feel procedural and exposed.

Direction benchmarkee : north Status : valid Updated place : Vaulted Corridor Intent :

move_north_generated_room Metadata source : llm Scene source : hybrid Rendered scene source :

generated_room_cache Turn fallback : not needed Prompt dump : 91.00 ms Command wall time : 13230.43 ms
Inference time : 9338.74 ms Preparation time : 12441.81 ms Compile estimate : 3103.07 ms Render time : 645.94
ms Prompt tokens : 799 Generated tokens : 387 Triangles : 628 Materials : 54 Scene lines : 20 Scene chars : 1689

Origin state :

generated/hybrid_scene_generation_benchmark/states/sally_port_checkpoint.origin.session.json

Result state :

generated/hybrid_scene_generation_benchmark/states/sally_port_checkpoint.result.session.json

Current generated room :

generated/hybrid_scene_generation_benchmark/states/sally_port_checkpoint.generated_room.json

Prompt : generated/hybrid_scene_generation_benchmark/prompts/sally_port_checkpoint.prompt.txt

Run stdout : generated/hybrid_scene_generation_benchmark/logs/sally_port_checkpoint.stdout.txt

Run stderr : generated/hybrid_scene_generation_benchmark/logs/sally_port_checkpoint.stderr.txt

Raw room JSON :

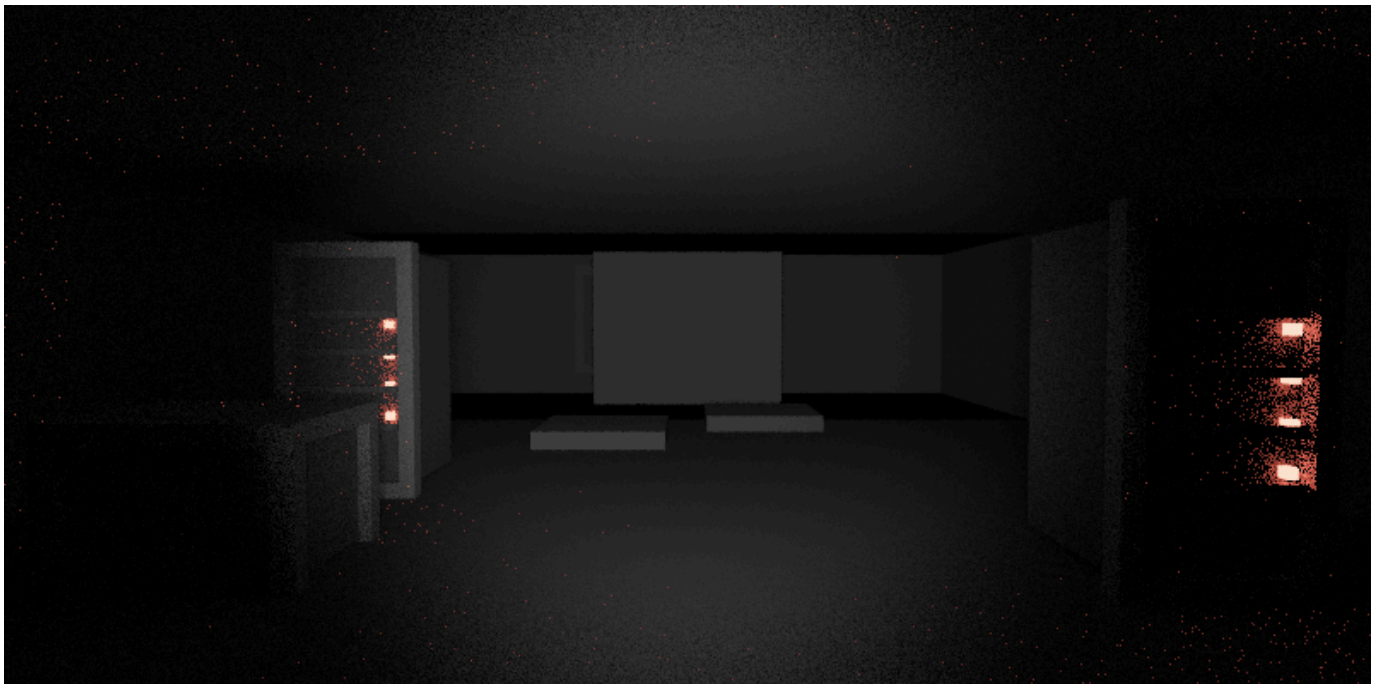
generated/hybrid_scene_generation_benchmark/responses/sally_port_checkpoint.raw.txt Normalized
room JSON :

generated/hybrid_scene_generation_benchmark/responses/sally_port_checkpoint.generated_room.js
on Compiled scene :

generated/hybrid_scene_generation_benchmark/scenes/sally_port_checkpoint.scene

Source room brief :

- location_id: unknown
- location_archetype: security checkpoint
- time_of_day: night
- visibility_level: low
- desert_state: windy
- interior_density: sparse
- alert_level: 2
- blocked_exits: south



Badge Vestibule

A bare vestibule buffers the exterior from the datacenter core. The only relief is a reader column, a maintenance hatch, and a wall placard with faded rules.

Direction benchmarkee : **north** Status : **valid** Updated place : **Access Node** Intent : **move_north_generated_room**
Metadata source : **11m** Scene source : **hybrid** Rendered scene source : **generated_room_cache** Turn fallback : **not needed** Prompt dump : **90.37 ms** Command wall time : **11406.21 ms** Inference time : **7549.01 ms** Preparation time : **10661.65 ms** Compile estimate : **3112.64 ms** Render time : **607.01 ms** Prompt tokens : **806** Generated tokens : **309** Triangles : **340** Materials : **30** Scene lines : **20** Scene chars : **1520**

Origin state :

generated/hybrid_scene_generation_benchmark/states/badge_vestibule.origin.session.json Result state : **generated/hybrid_scene_generation_benchmark/states/badge_vestibule.result.session.json**

Current generated room :

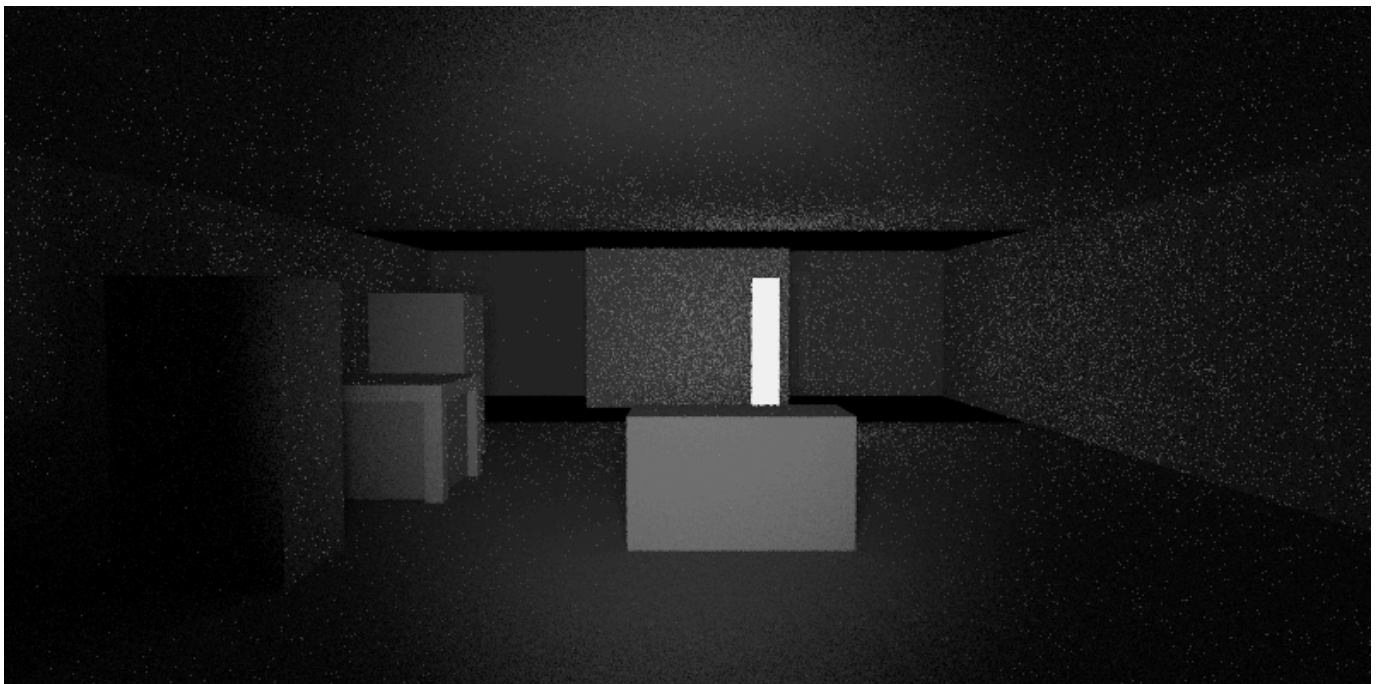
generated/hybrid_scene_generation_benchmark/states/badge_vestibule.generated_room.json Prompt : **generated/hybrid_scene_generation_benchmark/prompts/badge_vestibule.prompt.txt** Run stdout : **generated/hybrid_scene_generation_benchmark/logs/badge_vestibule.stdout.txt** Run stderr : **generated/hybrid_scene_generation_benchmark/logs/badge_vestibule.stderr.txt** Raw room JSON : **generated/hybrid_scene_generation_benchmark/responses/badge_vestibule.raw.txt** Normalized room JSON :

generated/hybrid_scene_generation_benchmark/responses/badge_vestibule.generated_room.json

Compiled scene : **generated/hybrid_scene_generation_benchmark/scenes/badge_vestibule.scene**

Source room brief :

- location_id: **unknown**
- location_archetype: **airlock vestibule**
- time_of_day: **night**
- visibility_level: **clear**
- desert_state: **still**
- interior_density: **sparse**
- alert_level: **1**
- blocked_exits: **west**



Primary Server Aisles

Dense aisles of racks run into darkness. A service cart, a floor hatch, and a side console break the repetition under red LEDs.

Direction benchmarkee : `north` Status : `valid` Updated place : `Backup Vault` Intent :
`move_north_generated_room` Metadata source : `llm` Scene source : `hybrid` Rendered scene source :
`generated_room_cache` Turn fallback : `not needed` Prompt dump : `91.71 ms` Command wall time : `13785.50 ms`
Inference time : `9187.81 ms` Preparation time : `12328.85 ms` Compile estimate : `3141.04 ms` Render time :
`1330.60 ms` Prompt tokens : `802` Generated tokens : `379` Triangles : `1062` Materials : `91` Scene lines : `22` Scene chars :
`1838`

Origin state :

`generated/hybrid_scene_generation_benchmark/states/server_aisles_dense.origin.session.json`

Result state :

`generated/hybrid_scene_generation_benchmark/states/server_aisles_dense.result.session.json`

Current generated room :

`generated/hybrid_scene_generation_benchmark/states/server_aisles_dense.generated_room.json`

Prompt : `generated/hybrid_scene_generation_benchmark/prompts/server_aisles_dense.prompt.txt` Run

stdout : `generated/hybrid_scene_generation_benchmark/logs/server_aisles_dense.stdout.txt` Run

stderr : `generated/hybrid_scene_generation_benchmark/logs/server_aisles_dense.stderr.txt` Raw

room JSON : `generated/hybrid_scene_generation_benchmark/responses/server_aisles_dense.raw.txt`

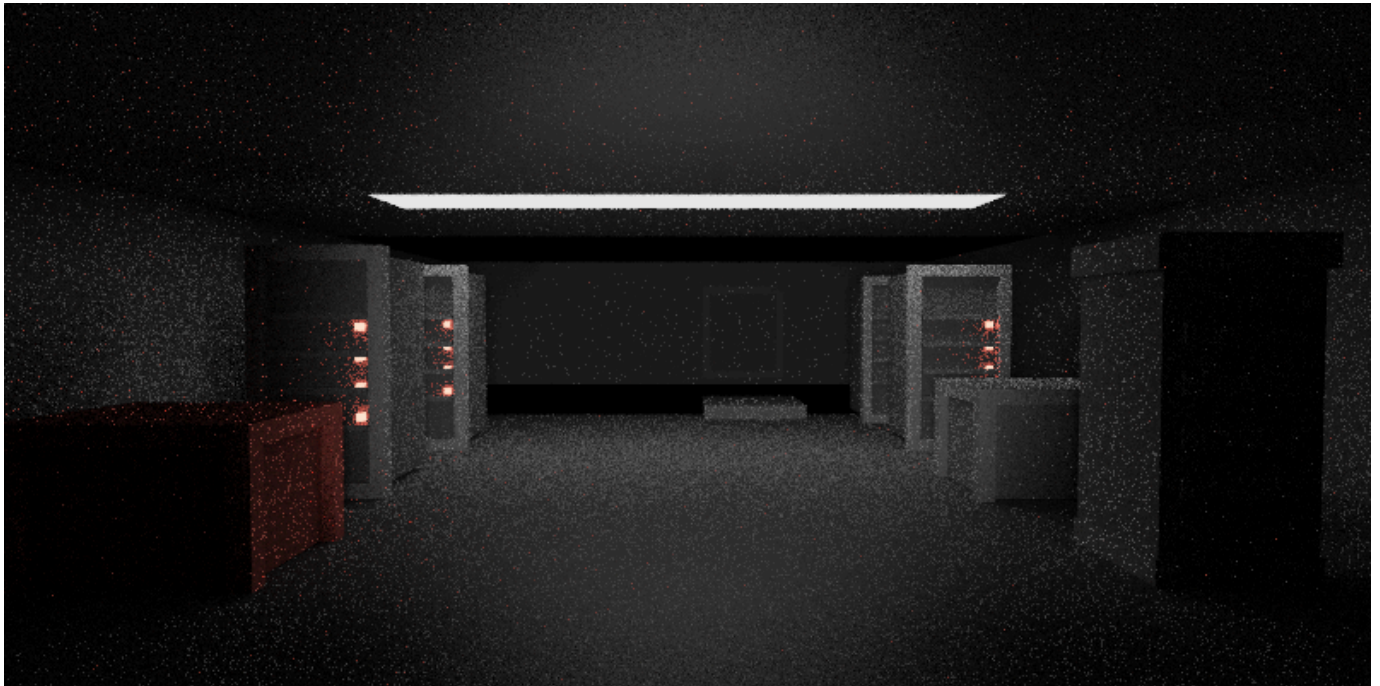
Normalized room JSON :

`generated/hybrid_scene_generation_benchmark/responses/server_aisles_dense.generated_room.json`

Compiled scene : `generated/hybrid_scene_generation_benchmark/scenes/server_aisles_dense.scene`

Source room brief :

- location_id: `server_aisles`
- location_archetype: `server aisles`
- time_of_day: `night`
- visibility_level: `low`
- desert_state: `still`
- interior_density: `dense`
- alert_level: `2`
- blocked_exits: `south`



Sector Control Hub

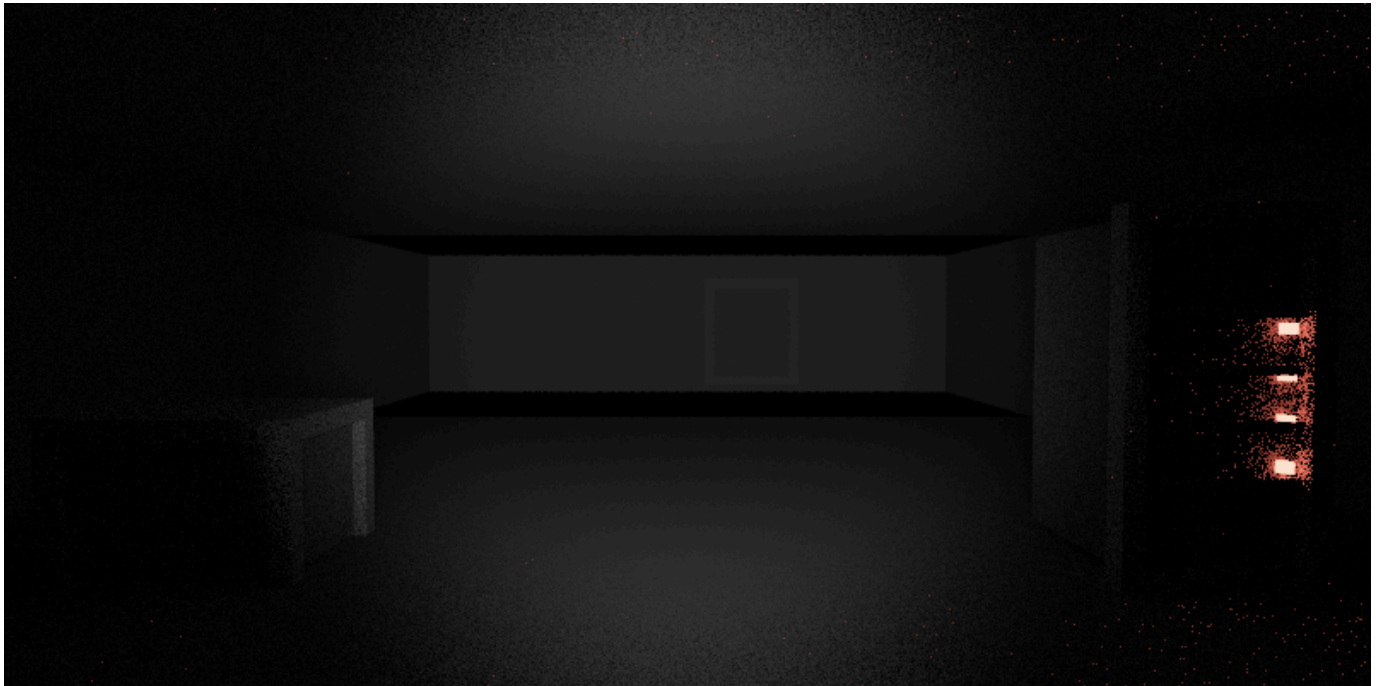
A compact control room opens off the aisles. One main console dominates the center while a locked cabinet and an access hatch wait at the margins.

Direction benchmarkee : **north** Status : **valid** Updated place : **Ventilation Grid** Intent : **move_north_generated_room** Metadata source : **llm** Scene source : **hybrid** Rendered scene source : **generated_room_cache** Turn fallback : **not needed** Prompt dump : **85.71 ms** Command wall time : **10861.51 ms** Inference time : **7077.20 ms** Preparation time : **10168.62 ms** Compile estimate : **3091.42 ms** Render time : **560.99 ms** Prompt tokens : **790** Generated tokens : **289** Triangles : **424** Materials : **37** Scene lines : **15** Scene chars : **1149**

Origin state : **generated/hybrid_scene_generation_benchmark/states/control_hub.origin.session.json**
Result state : **generated/hybrid_scene_generation_benchmark/states/control_hub.result.session.json**
Current generated room : **generated/hybrid_scene_generation_benchmark/states/control_hub.generated_room.json** Prompt : **generated/hybrid_scene_generation_benchmark/prompts/control_hub.prompt.txt** Run stdout : **generated/hybrid_scene_generation_benchmark/logs/control_hub.stdout.txt** Run stderr : **generated/hybrid_scene_generation_benchmark/logs/control_hub.stderr.txt** Raw room JSON : **generated/hybrid_scene_generation_benchmark/responses/control_hub.raw.txt** Normalized room JSON : **generated/hybrid_scene_generation_benchmark/responses/control_hub.generated_room.json** Compiled scene : **generated/hybrid_scene_generation_benchmark/scenes/control_hub.scene**

Source room brief :

- location_id: **unknown**
- location_archetype: **control room**
- time_of_day: **night**
- visibility_level: **clear**
- desert_state: **still**
- interior_density: **sparse**
- alert_level: **2**
- blocked_exits: **east, south**



Backup Vault

A chilled storage vault holds sealed pods and backup racks. A biometric pad, a wheeled crate, and a suspended service panel suggest expensive access.

Direction benchmarkee : **east** Status : **valid** Updated place : **Cable Trench** Intent : **move_east_generated_room**
Metadata source : **11m** Scene source : **hybrid** Rendered scene source : **generated_room_cache** Turn fallback : **not needed** Prompt dump : **90.75 ms** Command wall time : **12059.73 ms** Inference time : **8082.58 ms** Preparation time : **11261.46 ms** Compile estimate : **3178.88 ms** Render time : **668.25 ms** Prompt tokens : **802** Generated tokens : **329** Triangles : **676** Materials : **58** Scene lines : **19** Scene chars : **1528**

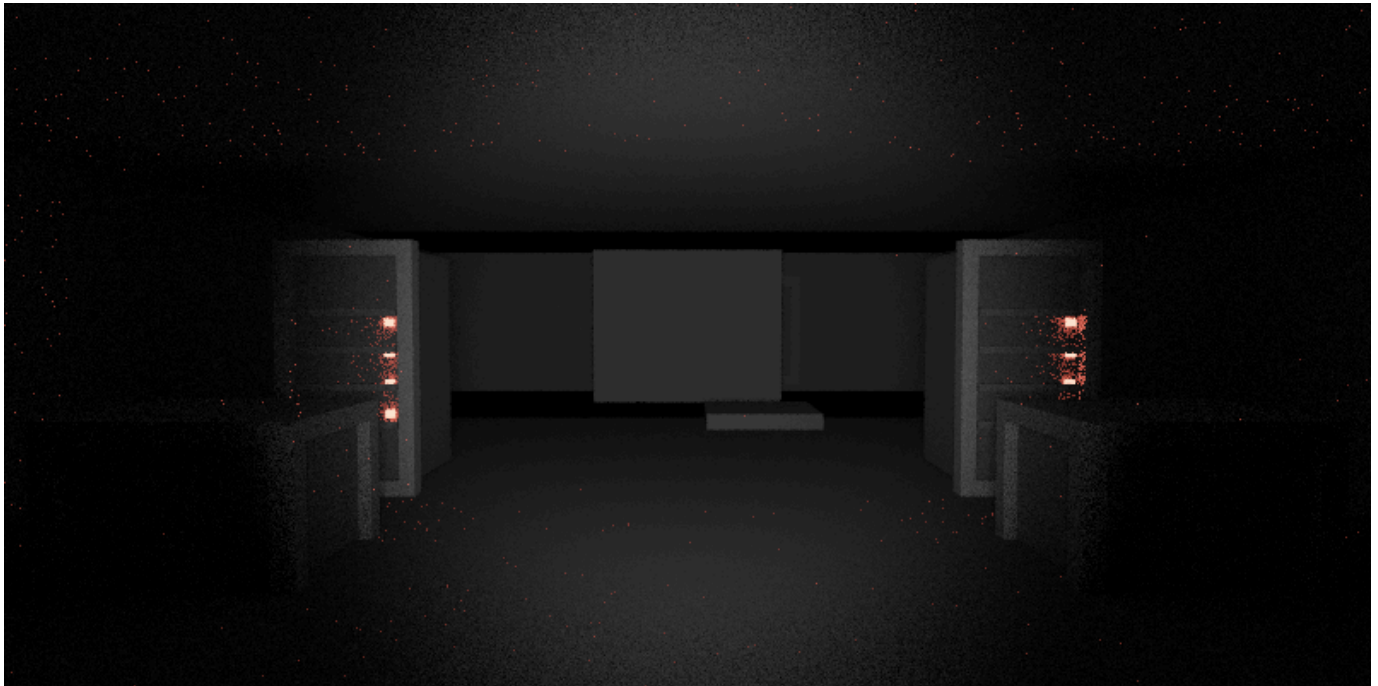
Origin state :

generated/hybrid_scene_generation_benchmark/states/backup_vault.origin.session.json Result state : **generated/hybrid_scene_generation_benchmark/states/backup_vault.result.session.json** Current generated room :

generated/hybrid_scene_generation_benchmark/states/backup_vault.generated_room.json Prompt : **generated/hybrid_scene_generation_benchmark/prompts/backup_vault.prompt.txt** Run stdout : **generated/hybrid_scene_generation_benchmark/logs/backup_vault.stdout.txt** Run stderr : **generated/hybrid_scene_generation_benchmark/logs/backup_vault.stderr.txt** Raw room JSON : **generated/hybrid_scene_generation_benchmark/responses/backup_vault.raw.txt** Normalized room JSON : **generated/hybrid_scene_generation_benchmark/responses/backup_vault.generated_room.json** Compiled scene : **generated/hybrid_scene_generation_benchmark/scenes/backup_vault.scene**

Source room brief :

- location_id: **unknown**
- location_archetype: **backup vault**
- time_of_day: **night**
- visibility_level: **low**
- desert_state: **still**
- interior_density: **dense**
- alert_level: **3**
- blocked_exits: **north, west**



Cooling Exchange Bay

Tall cooling blocks and pipe runs crowd an industrial chamber. A valve wheel, a drip tray, and a maintenance console provide the room's few obvious handles.

Direction benchmarkee : **north** Status : **valid** Updated place : **Ventilation Grid** Intent : **move_north_generated_room** Metadata source : **llm** Scene source : **hybrid** Rendered scene source : **generated_room_cache** Turn fallback : **not needed** Prompt dump : **89.65 ms** Command wall time : **12243.29 ms** Inference time : **8102.99 ms** Preparation time : **11222.40 ms** Compile estimate : **3119.41 ms** Render time : **890.40 ms** Prompt tokens : **803** Generated tokens : **333** Triangles : **520** Materials : **45** Scene lines : **17** Scene chars : **1382**

Origin state :

generated/hybrid_scene_generation_benchmark/states/cooling_exchange_bay.origin.session.json

Result state :

generated/hybrid_scene_generation_benchmark/states/cooling_exchange_bay.result.session.json

Current generated room :

generated/hybrid_scene_generation_benchmark/states/cooling_exchange_bay.generated_room.json

Prompt : **generated/hybrid_scene_generation_benchmark/prompts/cooling_exchange_bay.prompt.txt**

Run stdout : **generated/hybrid_scene_generation_benchmark/logs/cooling_exchange_bay.stdout.txt**

Run stderr : **generated/hybrid_scene_generation_benchmark/logs/cooling_exchange_bay.stderr.txt** Raw

room JSON : **generated/hybrid_scene_generation_benchmark/responses/cooling_exchange_bay.raw.txt**

Normalized room JSON :

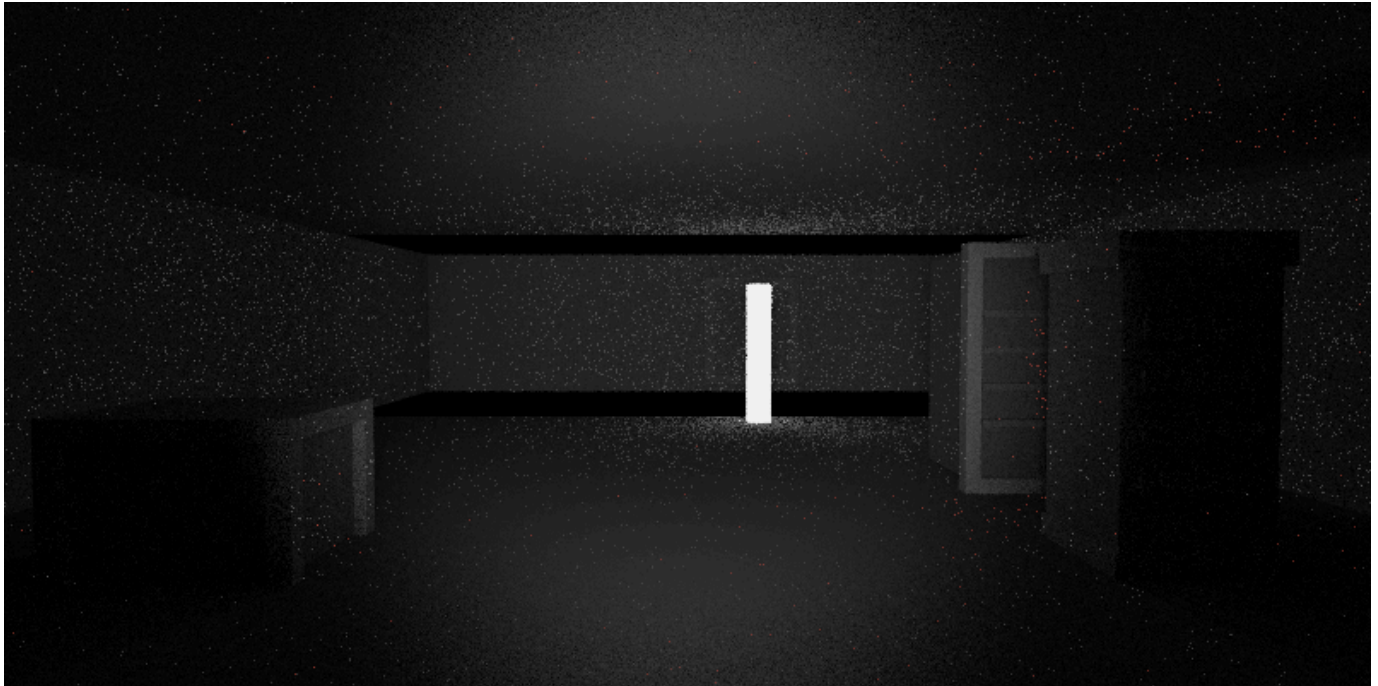
generated/hybrid_scene_generation_benchmark/responses/cooling_exchange_bay.generated_room.json

Compiled scene : **generated/hybrid_scene_generation_benchmark/scenes/cooling_exchange_bay.scene**

Source room brief :

- location_id: **unknown**
- location_archetype: **cooling plant bay**
- time_of_day: **night**
- visibility_level: **dusty**
- desert_state: **still**
- interior_density: **dense**

- alert_level: 2
- blocked_exits: west



Dust Loading Dock

A side loading bay opens toward the desert through a half-raised shutter. Pallets, a cargo crate, and a bent lamp post stage a tense threshold between inside and outside.

Direction benchmarkee : east Status : valid Updated place : Rusted Maintenance Tunnel Intent : move_east_generated_room Metadata source : llm Scene source : hybrid Rendered scene source : generated_room_cache Turn fallback : not needed Prompt dump : 90.23 ms Command wall time : 13542.20 ms Inference time : 9683.33 ms Preparation time : 12801.16 ms Compile estimate : 3117.83 ms Render time : 607.95 ms Prompt tokens : 813 Generated tokens : 401 Triangles : 436 Materials : 38 Scene lines : 16 Scene chars : 1348

Origin state :

generated/hybrid_scene_generation_benchmark/states/loading_dock_dust.origin.session.json Result state : generated/hybrid_scene_generation_benchmark/states/loading_dock_dust.result.session.json

Current generated room :

generated/hybrid_scene_generation_benchmark/states/loading_dock_dust.generated_room.json

Prompt : generated/hybrid_scene_generation_benchmark/prompts/loading_dock_dust.prompt.txt Run

stdout : generated/hybrid_scene_generation_benchmark/logs/loading_dock_dust.stdout.txt Run stderr :

generated/hybrid_scene_generation_benchmark/logs/loading_dock_dust.stderr.txt Raw room JSON :

generated/hybrid_scene_generation_benchmark/responses/loading_dock_dust.raw.txt Normalized room JSON :

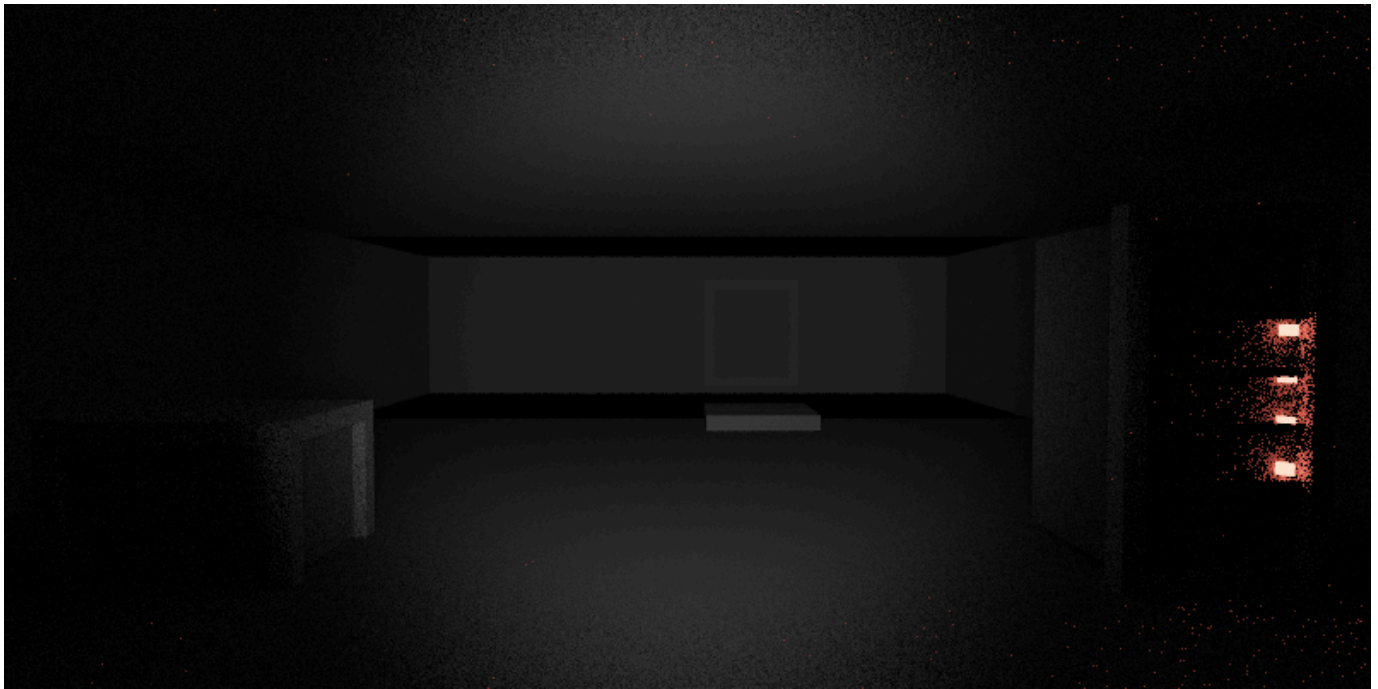
generated/hybrid_scene_generation_benchmark/responses/loading_dock_dust.generated_room.json

Compiled scene : generated/hybrid_scene_generation_benchmark/scenes/loading_dock_dust.scene

Source room brief :

- location_id: unknown
- location_archetype: loading dock
- time_of_day: dusk
- visibility_level: dusty
- desert_state: dusty

- interior_density: **sparse**
- alert_level: **2**
- blocked_exits: **north**



Roof Watch

The datacenter roof becomes a parapet walk above the desert. A beacon mast, a hatch door, and a field scope hold the line against the horizon.

Direction benchmarkee : **north** Status : **valid** Updated place : **Beacon Tower** Intent :
move_north_generated_room Metadata source : **llm** Scene source : **hybrid** Rendered scene source :
generated_room_cache Turn fallback : **not needed** Prompt dump : **91.40 ms** Command wall time : **12315.88 ms**
Inference time : **7900.49 ms** Preparation time : **11002.84 ms** Compile estimate : **3102.35 ms** Render time :
1176.02 ms Prompt tokens : **806** Generated tokens : **324** Triangles : **894** Materials : **77** Scene lines : **21** Scene chars :
1748

Origin state :

generated/hybrid_scene_generation_benchmark/states/roof_watch_dusk.origin.session.json Result
state : **generated/hybrid_scene_generation_benchmark/states/roof_watch_dusk.result.session.json**

Current generated room :

generated/hybrid_scene_generation_benchmark/states/roof_watch_dusk.generated_room.json Prompt :
generated/hybrid_scene_generation_benchmark/prompts/roof_watch_dusk.prompt.txt Run stdout :
generated/hybrid_scene_generation_benchmark/logs/roof_watch_dusk.stdout.txt Run stderr :
generated/hybrid_scene_generation_benchmark/logs/roof_watch_dusk.stderr.txt Raw room JSON :
generated/hybrid_scene_generation_benchmark/responses/roof_watch_dusk.raw.txt Normalized room
JSON :

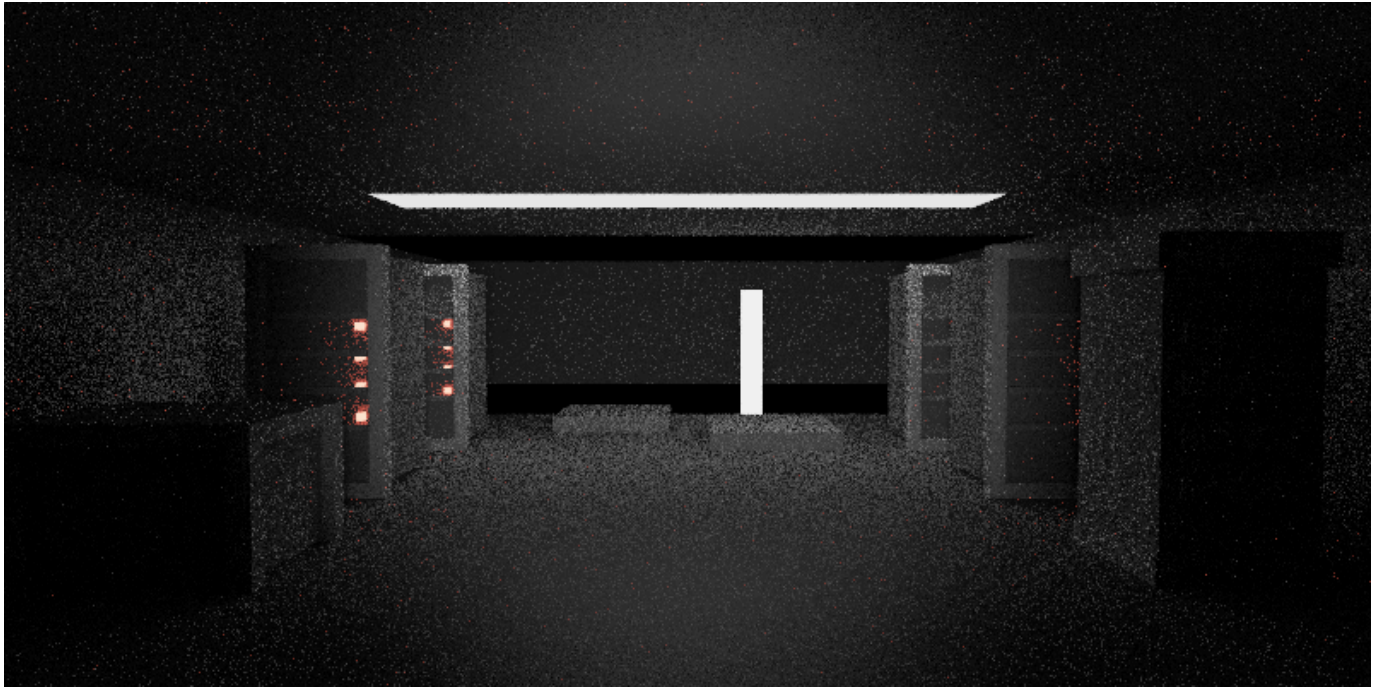
generated/hybrid_scene_generation_benchmark/responses/roof_watch_dusk.generated_room.json

Compiled scene : **generated/hybrid_scene_generation_benchmark/scenes/roof_watch_dusk.scene**

Source room brief :

- location_id: **roof_watch**
- location_archetype: **roof parapet**
- time_of_day: **dusk**

- visibility_level: **clear**
- desert_state: **windy**
- interior_density: **sparse**
- alert_level: **1**
- blocked_exits: **south**



Parapet Stairhead

A concrete stairhead interrupts the roof line. The landing is tight, with a latch box, a cable spool, and a narrow gate looking over the darkening desert.

Direction benchmarkee : **north** Status : **valid** Updated place : **Ventilation Shaft** Intent : **move_north_generated_room** Metadata source : **llm** Scene source : **hybrid** Rendered scene source : **generated_room_cache** Turn fallback : **not needed** Prompt dump : **91.99 ms** Command wall time : **12319.47 ms** Inference time : **8223.88 ms** Preparation time : **11470.38 ms** Compile estimate : **3246.50 ms** Render time : **701.68 ms** Prompt tokens : **818** Generated tokens : **336** Triangles : **508** Materials : **44** Scene lines : **17** Scene chars : **1354**

Origin state :

generated/hybrid_scene_generation_benchmark/states/stairhead_parapet.origin.session.json Result state : **generated/hybrid_scene_generation_benchmark/states/stairhead_parapet.result.session.json**

Current generated room :

generated/hybrid_scene_generation_benchmark/states/stairhead_parapet.generated_room.json

Prompt : **generated/hybrid_scene_generation_benchmark/prompts/stairhead_parapet.prompt.txt** Run

stdout : **generated/hybrid_scene_generation_benchmark/logs/stairhead_parapet.stdout.txt** Run stderr :

generated/hybrid_scene_generation_benchmark/logs/stairhead_parapet.stderr.txt Raw room JSON :

generated/hybrid_scene_generation_benchmark/responses/stairhead_parapet.raw.txt Normalized room JSON :

generated/hybrid_scene_generation_benchmark/responses/stairhead_parapet.generated_room.json

Compiled scene : **generated/hybrid_scene_generation_benchmark/scenes/stairhead_parapet.scene**

Source room brief :

- location_id: **unknown**
- location_archetype: **roof stairhead**

- time_of_day: **dusk**
- visibility_level: **low**
- desert_state: **windy**
- interior_density: **sparse**
- alert_level: **1**
- blocked_exits: **east, south**

